

1.  $m = 0.4 \text{ Kg}$   $u = +4 \text{ m/s}$

$\dot{J} = 32.4 \text{ N/s}$   $\Delta t = 27 \times 10^{-3} \text{ s}$

a)  $P_o = mu = 0.4 \times 4 = 5.6 \text{ Kg m/s}$

$\Delta p = -32.4 \text{ Kg m/s}$

$P_f = P_o + \Delta p = 5.6 - 32.4 = -26.8 \text{ Kg m/s}$

$v = \frac{P_f}{m} = \frac{-26.8}{0.4} = -67 \text{ m/s}$  ~~the~~  $-ve \times$  direction

b)  $F = \frac{J}{\Delta t} = \frac{32.4}{0.027} = 1.2 \times 10^3 \text{ N}$   $-ve \times$

2.  $m = 10 \text{ Kg}$   $\vec{d} = 20 \text{ m } \hat{i} - 16 \text{ m } \hat{j}$   $\vec{F} = 210 \text{ N } \hat{i} - 150 \text{ N } \hat{j}$   $u = 0$

(a)  $W = \vec{F} \cdot \vec{d} = (210 \text{ N } \hat{i} - 150 \text{ N } \hat{j}) \cdot (20 \text{ m } \hat{i} - 16 \text{ m } \hat{j})$   
 $= (210)(20) + (-150)(-16)$   
 $= 4200 + 2400 = 6600 \text{ J}$

$W = 6600 \text{ J}$

(b) Work-Energy Theorem  $W = \Delta K = \frac{1}{2} m v^2 - \frac{1}{2} m u^2$

$6600 = \frac{1}{2} (10) v^2$

$v = \sqrt{\frac{6600 \times 2}{10}} = \sqrt{1320} \approx 36.3 \text{ m/s}$

final velocity  $= 36.3 \text{ m/s}$

3. The distance from the square's centre to each corner is half the diagonal

$$r = \frac{a\sqrt{2}}{2}$$

$$= \frac{a}{\sqrt{2}}$$

$$I = \sum mr^2 = 4m \left( \frac{a}{\sqrt{2}} \right)^2$$

$$= 4 \cdot m \left( \frac{a^2}{2} \right)$$

$$= 2ma^2$$

$$\therefore, I = 2ma^2$$

$\omega_0 = 20 \text{ rad/s}$     $\omega = 0 \text{ rad/s}$     $t = 4 \text{ s}$     $I = 0.2 \text{ kg/m}^2$   
 a)  $\alpha = \frac{\omega - \omega_0}{t}$

$$= \frac{0 - 20}{4} = -5 \text{ rad/s}^2$$

angular deceleration =  $5 \text{ rad/s}^2$

b)  $\theta = \omega_0 t + \frac{1}{2} \alpha t^2$

$$= 20(2.0) + \frac{1}{2}(-5)(2)^2$$

$$= 40 - 10$$

$$= 30 \text{ rad}$$

So the angle rotated in the first 2 sec  
 $\Rightarrow 30 \text{ rad}$

$$\begin{aligned}
 5) \quad m &= 2.4 \text{ kg} \quad \theta = 37^\circ \quad u = 3.8 \text{ m/s} \quad \mu_k = 0.3 \\
 a &= -(g \sin \theta + \mu_k g \cos \theta) \\
 &= -(9.8 \sin 37^\circ + 0.3 \times 9.8 \cos 37^\circ) \\
 &= -(9.8 \times 0.601) + (0.3 \times 9.8 \times 0.8 \cos 37^\circ) \\
 &= -(5.890 + 2.352) \\
 &= -8.242 \text{ m/s}^2 \\
 |a| &= 8.242 \text{ m/s}^2
 \end{aligned}$$

$$v^2 = u^2 + 2as \quad v = 0$$

$$0 = u^2 - 2|a|s$$

$$0 = (3.8)^2 - 2 \times 8.242 \text{ s}$$

$$16.4848 = 16.44$$

$$s = \frac{16.44}{16.484} \approx 0.88 \text{ m}$$

Distance travelled before stopping  
 $\approx 0.88 \text{ m}$  up the incline



6)  $F = 30 \text{ N}$   $m = 10 \text{ kg}$   $u = 0$   $S = 24 \text{ m}$

i)  $W = F \cdot s = 30 \cdot 24 = 720 \text{ J}$   
 $K_{\text{final}} = 720 \text{ J}$

ii)  $K = \frac{1}{2} m v^2 = 720$

$\Rightarrow v = \sqrt{\frac{720 \cdot 2}{m}} = \sqrt{\frac{1440}{10}} = \sqrt{144}$

$v = 12 \text{ m/s}$

iii)  $P_{\text{final}} = F \cdot v_{\text{final}} = 30 \cdot 12 = 360 \text{ W}$

$P = \frac{W}{t}$   $a = \frac{F}{m} = \frac{30}{10} = 3 \text{ m/s}^2$

$S = \frac{1}{2} a t^2 \Rightarrow 24 = \frac{1}{2} \cdot 3 \cdot t^2$

$t = \sqrt{\frac{24 \cdot 2}{3}} = \sqrt{16}$

$t = 4 \text{ s}$

$\hookrightarrow P = \frac{720}{4} = 180 \text{ W}$

$$7) \omega_0 = 24 \text{ rad/s} \quad \alpha_1 = 30 \text{ rad/s}^2 \quad \theta_2 = 432 \text{ rad}$$

$$\theta_1 = \omega_0 t + \frac{1}{2} \alpha_1 t^2 = 24(2) + \frac{1}{2} (30)(2)^2 = 48 + 60 = 108 \text{ rad}$$

$$\theta_2 = \frac{1}{2} (\omega_1 + 0) t_2 = 42 t_2$$

$$t_2 = \frac{432}{42} \approx 10.286 \text{ s}$$

$$0 = \omega_1 + \alpha_2 t_2$$

$$\alpha_2 = \frac{-\omega_1}{t_2} = \frac{-84}{10.286} \approx -8.17 \text{ rad/s}^2$$

$$\alpha_2 = \frac{-\omega_1^2}{2\theta_2} = \frac{-84^2}{2 \cdot 432} \approx -8.17 \text{ rad/s}^2$$

$$i) \theta_{\text{total}} = \theta_1 + \theta_2 = 108 + 432 = 540 \text{ rad}$$

$$ii) t_{\text{stop}} = 2 + t_2 = 2 + 10.286 = 12.286 \text{ s}$$

$$iii) \text{ angular deceleration } \alpha_2 \approx -8.17 \text{ rad/s}^2$$

8. Initially the object is stationary, so  
total momentum = 0

$$m_A v_A + m_B v_B = 0$$

$$m_A v_A = m_B v_B$$

i)  $v_B = \frac{m_A v_A}{m_B}$

ii)  $K_A = \frac{1}{2} m_A v_A^2$   
 $K_B = \frac{1}{2} m_B v_B^2$

$$v_B = \frac{m_A v_A}{m_B}$$

$$K_B = \frac{1}{2} m_B \left( \frac{m_A v_A}{m_B} \right)^2 = \frac{1}{2} \frac{m_A^2 v_A^2}{m_B}$$

$$\frac{K_A}{K_B} = \frac{\frac{1}{2} m_A v_A^2}{\frac{1}{2} \frac{m_A^2 v_A^2}{m_B}}$$

$$= \frac{m_A}{\frac{m_A^2}{m_B}} = \frac{m_B}{m_A}$$

$$\frac{K_A}{K_B} = \frac{m_B}{m_A}$$



4)  $T = 0.300 \text{ s}$     $A = 6.00 \text{ cm}$     $t = 0 \quad x(0) = +6.00 \text{ cm}$   
~~Because the object starts at~~

$$x(t) = 6.00 \cos(\omega t)$$

when  $x = -1.50 \text{ cm}$

$$-1.5 = 6 \cos(\omega t)$$

$$\cos(\omega t) = \frac{-1.5}{6} = -0.25$$

$$\omega t = \cos^{-1}(-0.25)$$

$$\cos^{-1}(-0.25) \approx 1.823 \text{ rad}$$

$$t = \frac{1.823}{\omega} = \frac{1.823}{20.94} \approx 0.087 \text{ s}$$

The object takes about  $0.087 \text{ s}$  to move from  
 $x = +6.00 \text{ cm}$  to  $x = -1.50 \text{ cm}$



16.  $m = 65 \text{ kg}$

i)  $T = \frac{43}{8} = 5.375 \text{ s}$

$$T = 2\pi \sqrt{\frac{m}{K}} \Rightarrow K = \frac{4\pi^2 m}{T^2}$$

$$K = \frac{4\pi^2 \cdot 65}{(5.375)^2} = 8.89 \times 10 \text{ N/m} = 88.9 \text{ N/m}$$

Spring stiffness constant  $k \approx 89 \text{ N/m}$

ii)  $mg = kx_{\text{eq}} \Rightarrow x = \frac{mg}{k} = \frac{65 \times 9.8}{88.9} \approx 7.17$

The rest position is 25 m below the bridge which equals the cord's unstretched length plus the equilibrium extension

$$l_0 = 25.0 - x = 25 - 7.17 = 17.83 \text{ m}$$

$$l_0 \approx 17.8 \text{ m}$$

Unstretched length  $l_0 \approx 17.8 \text{ m}$

$$A = 3 \text{ m}$$

$$\omega = \pi \text{ rad/s}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\pi} = 2 \text{ s}$$

$$v_{\text{max}} = A\omega = 3\pi = 3\pi \text{ m/s} \approx 9.425 \text{ m/s}$$

~~at t = 2 s~~

$$v(t) = \frac{dx}{dt} = 3\pi \cos\left(\pi t + \frac{\pi}{3}\right)$$

$$\begin{aligned} \text{at } t = 2 \text{ s} \quad v(2) &= 3\pi \cos\left(2\pi + \frac{\pi}{3}\right) \\ &= 3\pi \cos\left(\frac{\pi}{3}\right) \\ &= 3\pi \cdot \frac{1}{2} \\ &= \frac{3\pi}{2} \text{ m/s} \approx 4.712 \text{ m/s} \end{aligned}$$

12:

$$E_i = E_f$$

$$mgh_i + \frac{1}{2}mv_i^2 = mgh_f + \frac{1}{2}mv_f^2$$

$$mgR + 0 = 0 + \frac{1}{2}mv^2$$

$$v_i^2 = 2gR$$

$$v = \sqrt{2gR}$$

for particle of the right

$$v_2 = \sqrt{2gR}$$

$$U_1 = \sqrt{2gR}$$

$$V_1 = \sqrt{2gR}$$

$$U_2 = \sqrt{2gR}$$

$$V_2 = \sqrt{2gR}$$

maximum height

$$mgh_i + \frac{1}{2}mv_i^2 = mgh + \frac{1}{2}mv^2 \text{ final}$$

$$0 + \frac{1}{2}m(\sqrt{2gR})^2 = mgH_{\text{max}} + 0$$

$$\frac{1}{2}m(2gR) = mgH_{\text{max}} \Rightarrow H = R$$



$$H = R$$

maximum height reached by masses  
after collision is  $R$

when they stick together (in elastic collision)

$$m + (\sqrt{2gR}) + m(-\sqrt{2gR}) = \cancel{V} 2m$$

$$2mV = 0$$

$$V = 0$$

13. initial angular speed =  $500 \text{ rev/min} = \frac{500}{60} = 8.33 \text{ rev/s}$   
 final angular speed =  $200 \text{ rev/min} = \frac{200}{60} = 3.33 \text{ rev/s}$   
 $t = 4 \text{ s}$

a)  $\alpha = \frac{\omega_f - \omega_i}{t} = \frac{3.33 - 8.33}{4} = -1.25 \text{ rev/s}^2$

no of revolution in 4s

$$\theta = \omega t = \left( \frac{\omega_i + \omega_f}{2} \right) t = \left( \frac{8.33 + 3.33}{2} \right) 4$$

$$\theta = 23.33 \text{ rev}$$

b)  $200 \text{ rev/min} = \cancel{3.33} 3.33 \text{ rev/s}$

time from 3.33 to rest with same  $\alpha$

$$t_{\text{extra}} = \frac{-\omega_f}{\alpha} = \frac{-3.33}{-1.25} \approx 2.67 \text{ s}$$

additional time to stop = 2.67s

14) Same as question (1)

$$(15) \quad \omega_f^2 = \omega_i^2 + 2\alpha\theta$$

$$\omega_f = 0$$

$$\therefore 0 = \omega_i^2 + 2\alpha\theta \Rightarrow \theta = -\frac{\omega_i^2}{2\alpha}$$

since  $\alpha < 0$ , the angle comes out +ve

Initial speed =  $\omega_1$

$$\theta_1 = 1.00 \text{ rev}$$

$$\theta_1 = -\frac{\omega_1^2}{2\alpha}$$

Initial speed =  $\omega_3 = 3\omega_1$

$$\theta_3 = -\frac{\omega_3^2}{2\alpha} = -\frac{(3\omega_1)^2}{2\alpha} = \frac{9\omega_1^2}{2\alpha}$$

$$\text{Since } -\frac{\omega_1^2}{2\alpha} = \theta_1 = 1 \text{ rev}$$

$$\theta_3 = 9\theta_1 = 9(1.00)$$

$$\theta_3 = 9.00 \text{ revolutions}$$

If the initial angular speed is tripled, the blade requires 9 revolutions to come to rest under the same constant angular deceleration.



16.  $\theta(t) = \gamma t + \beta t^3$      $\gamma = 0.400 \text{ rad/s}$      $\beta = 0.0120 \text{ rad/s}^3$

(a)  $\omega(t) = \frac{d\theta}{dt} = \gamma + 3\beta t^2$   
 $= 0.400 + 3(0.0120)t^2$   
 $= 0.400 + 0.0360t^2 \text{ rad/s}$

b) at  $t = 0$   
 $\omega(0) = 0.400 + 0.0360(0)^2$   
 $= 0.400 \text{ rad/s}$

c) at  $t = 5 \text{ s}$   
 $\omega(5) = 0.400 + 0.0360(5)^2$   
 $= 0.400 + 0.0360(25)$   
 $= 0.400 + 0.900$   
 $= 1.300 \text{ rad/s}$

$\omega_{av} = \frac{\Delta\theta}{\Delta t} = \frac{\theta(5) - \theta(0)}{5}$   
 $= \frac{\gamma(5) + \beta(5)^3 - 0}{5} = \frac{0.400(5) + 0.0120(125)}{5}$   
 $= \frac{3.50}{5} = 0.700 \text{ rad/s}$

$$\frac{\omega(0) + \omega(5)}{2} = \frac{0.400 + 1.300}{2}$$

$$= \frac{1.700}{2} = 0.850 \text{ rad/s}$$

$$\frac{\omega(0) + \omega(5)}{2} = 0.850 \text{ rad/s} \neq 0.700 \text{ rad/s}$$

they are n't equal because angular velocity does not increase linearly with time it increases quadratically  $\omega(t) = \gamma + 3\beta t^2$

When the rate of change is non linear, the average value of a function over an interval is not equal to the arithmetic mean of its endpoint values.

$$\omega_{av} = \frac{1}{T} \int_0^T \omega(t) dt \neq \frac{\omega(0) + \omega(T)}{2}$$

unless  $\omega(t)$  is a straight-line function of time. Because  $\omega(t)$  curves upward (quadratic), it spends more time at lower angular velocities early on, so the time average is smaller than the end point average.